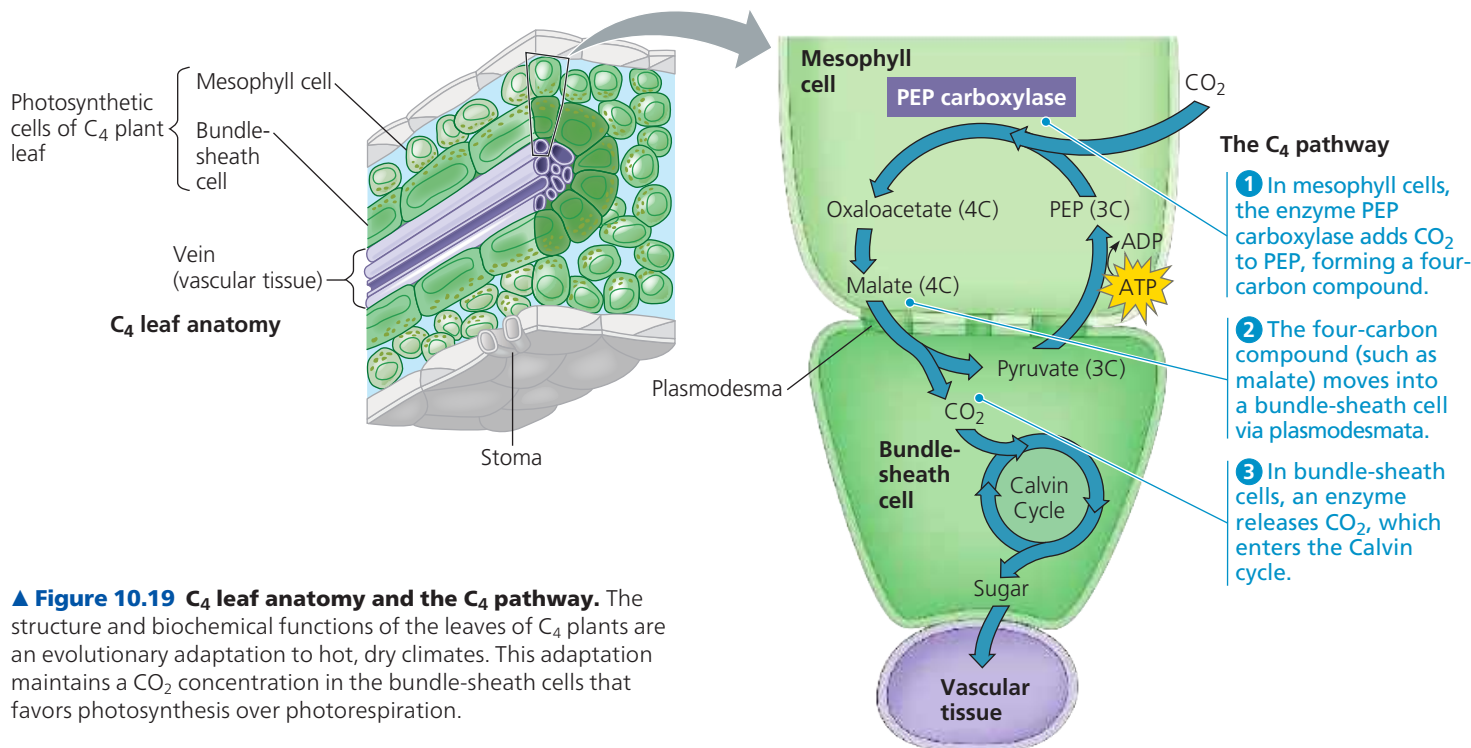




▲ **Figure 10.19 C₄ leaf anatomy and the C₄ pathway.** The structure and biochemical functions of the leaves of C₄ plants are an evolutionary adaptation to hot, dry climates. This adaptation maintains a CO₂ concentration in the bundle-sheath cells that favors photosynthesis over photorespiration.

phosphoenolpyruvate (PEP), forming the four-carbon product oxaloacetate; it does this even under conditions of lower CO₂ concentration and relatively higher O₂ concentration.

- 2 After the CO₂ is fixed in the mesophyll cells, the four-carbon products (malate in the example shown in Figure 10.19) are exported to bundle-sheath cells through plasmodesmata (see Figure 6.29).
- 3 Within the bundle-sheath cells, an enzyme releases CO₂ from the four-carbon compounds; the CO₂ is re-fixed into organic material by rubisco and the Calvin cycle. The same reaction regenerates pyruvate, which is transported to mesophyll cells. There, ATP is used to convert pyruvate to PEP, which can then accept addition of another CO₂, allowing the reaction cycle to continue. This ATP can be thought of, in a sense, as the “price” of concentrating CO₂ in the bundle-sheath cells. To generate this extra ATP, bundle-sheath cells carry out cyclic electron flow, the process described earlier in this chapter (see Figure 10.15). In fact, these cells contain PS I but no PS II, so cyclic electron flow is their only photosynthetic mode of generating ATP.

In effect, the mesophyll cells of a C₄ plant pump CO₂ into the bundle-sheath cells, keeping the CO₂ concentration in those cells high enough for rubisco to bind CO₂ rather than O₂. The cyclic series of reactions involving PEP carboxylase and the regeneration of PEP can be thought of as an ATP-powered pump that concentrates CO₂ where it can be fixed. In this way, C₄ photosynthesis spends ATP energy to minimize photorespiration and enhance sugar production. This adaptation is especially advantageous in hot regions with

intense sunlight, and it is in such environments that C₄ plants evolved and thrive today.

The concentration of CO₂ in the atmosphere has drastically increased since the Industrial Revolution began in the 1800s, and it continues to rise today due to human activities such as the burning of fossil fuels (see Concept 1.1). The resulting global climate change, including an increase in average temperatures around the planet, may have far-reaching effects on plant species. Scientists are concerned that increasing CO₂ concentration and temperature may affect C₃ and C₄ plants differently, thus changing the relative abundance of these species in a given plant community.

Which type of plant would stand to gain more from increasing CO₂ levels? Recall that in C₃ plants, the binding of O₂ rather than CO₂ by rubisco leads to photorespiration, lowering the efficiency of photosynthesis. C₄ plants overcome this problem by concentrating CO₂ in the bundle-sheath cells at the cost of ATP. Rising CO₂ levels should benefit C₃ plants by lowering the amount of photorespiration that occurs. At the same time, rising temperatures have the opposite effect, increasing photorespiration. (Other factors such as water availability may also come into play.) In contrast, many C₄ plants could be largely unaffected by increasing CO₂ levels or temperature. Researchers have investigated aspects of this question in several studies; you can work with data from one such experiment in the **Scientific Skills Exercise**. In different regions, the particular combination of CO₂ concentration and temperature is likely to alter the balance of C₃ and C₄ plants in varying ways. The effects of such a widespread and variable change in community structure are unpredictable and thus a cause of legitimate concern.